# Probability
# Lecture 5: Random Vectors

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Hanoi, 10/2024

# Content


1. Random vectors
2. Independent Random Variables
3. Joint Expectation, Covariance, Correlation
4. Homework

# Previous lecture

- Recall that:
+ $(\Omega, \mathcal{F}, \mathbb{P})$ a probability space with $\Omega$: set of all possible outcomes.
+ A random variable is a measurable function $X : (\Omega, \mathcal{F}, \mathbb{P}) \to (\mathbb{R}, \mathcal{B}(\mathbb{R}))$, i.e we consider the value of the outcome.
+ Discrete random variables: the set of possible values is finite or countable.
+ Continuous random variables: the Cumulative Distribution Function (CDF) is differentiable; its derivative is Probability Density Function (PDF).

# Functions of Random Variables

- Given a random variable $X$. For a function $g : \mathbb{R} \to \mathbb{R}$, define $Y = g(X)$.

$$F_Y(y) = \mathbb{P}(Y \leq y) = \mathbb{P}(g(X) \leq y).$$

- If $g$ is monotonically increasing, then

$$F_Y(y) = \mathbb{P}(g(X) \leq y) = \mathbb{P}(X \leq g^{-1}(y)) = F_X(g^{-1}(y)).$$

+ $X$ has density function $f_X(x)$

$$F_Y(y) = \int_{-\infty}^{g^{-1}(y)} f_X(x)dx.$$

+ Hence $Y$ has density function $f_Y(t)$

$$f_Y(y) = \frac{d}{dy}F_Y(y) = \frac{d}{dy}\int_{-\infty}^{g^{-1}(y)} f_X(x)dx = f_X(g^{-1}(y))\frac{dg^{-1}(y)}{dy}.$$

# Example

- Let $X$ be a random variable with density function

$$f_X(x) = ae^x e^{-ae^x}.$$

Let $Y = e^X$, and find $f_Y(y)$.
+ Hint: $g(x) = e^x$ then $g^{-1}(y) = \log y$.

$$f_Y(y) = f_X(g^{-1}(y))\frac{dg^{-1}(y)}{dy} = ae^{\log y}e^{-ae^{\log y}}(\log y)' = ae^{-ay}.$$

$\Rightarrow Y \sim Exp(a)$.

- Consider $X \sim U[-1,1]$. Define $Y = X^2$. Find the distribution of $Y$.

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# Random vectors

## Definition

Consider a set of random variables $X_1, \ldots, X_n$ defining on a common probability space $(\Omega, \mathcal{F}, \mathbb{P})$. We can define the random vector

$$X = (X_1, \ldots, X_n) : \ \Omega \to \mathbb{R}^n$$
$$\omega \mapsto X(\omega) = (X_1(\omega, \ldots, X_n(\omega)).$$

- In other ways, we can define a random vector as a random element taking value in $\mathbb{R}^n$ and define the marginal entries $X_1, \ldots, X_n$.
- We consider the space $\Omega$ through many directions.
+ $\Omega$ is a class; $X_i$: grade of Maths, Physics, Literature, ...
+ $\Omega$ is a population; $X_i$: height, weight, ...
- We will consider two special kinds: discrete and continuous

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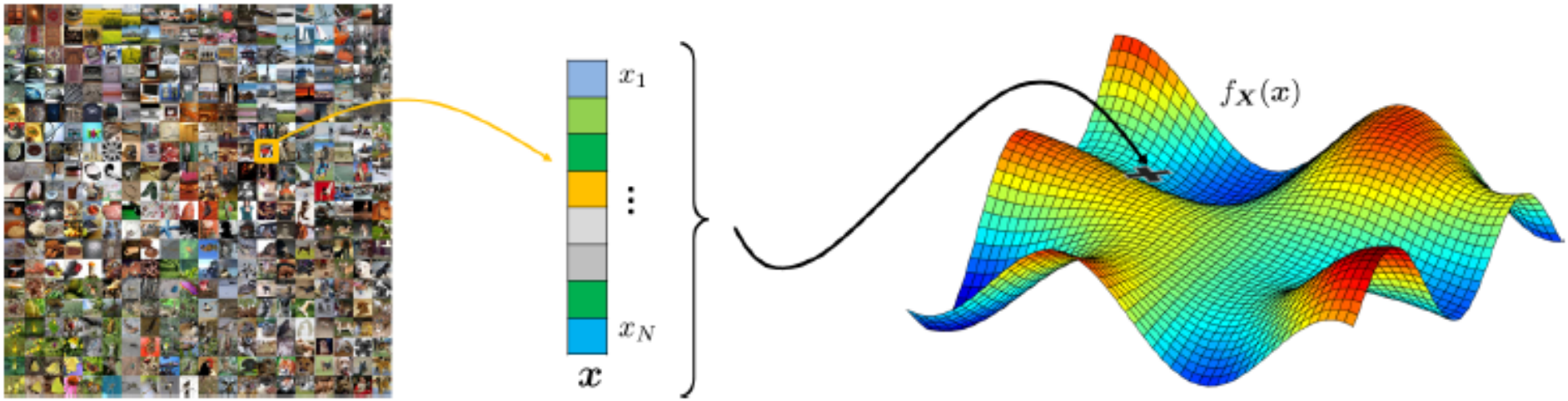



Figure: ImageNet dataset of 224$x$224$x$3 images.

# Discrete random vectors

- Joint distribution table

| $X \setminus Y$ | $y_1$ | $\cdots$ | $y_m$ |
|---|---|---|---|
| $x_1$ | $p_{11}$ | $\vdots$ | $p_{1m}$ |
| $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$ |
| $x_n$ | $p_{n1}$ | $\vdots$ | $p_{nm}$, |

where $p_{ij} = \mathbb{P}(\{X = x_i\} \cap \{Y = y_j\})$: **joint distribution**.

# Discrete random vectors

## Theorem

*We have*

- $\sum_{i,j} p_{ij} = 1.$
- *For any* $i = 1, \ldots, n,$

$$\sum_{j=1}^{m} p_{ij} = \mathbb{P}(X = x_i).$$

- *For any* $j = 1, \ldots, m,$

$$\sum_{i=1}^{n} p_{ij} = \mathbb{P}(Y = y_j).$$

# Example

- An urn contains 3 red balls, 4 blue balls and 2 yellow balls. Pick up randomly 2 ball from that urn and denote $X$ and $Y$ the number of red and yellow balls in the 2 balls, respectively.

1. Make the joint distribution table of $X$ and $Y$.
2. Find the distribution of $Z = XY$.

| $X \setminus Y$ | 0 | 1 | 2 |
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| $Z$ | 0 | 1 |
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| $Z$ | 0 | 1 |
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# Joint distribution function

## Definition

Given a random vector $X = (X_1, \ldots, X_n)$. We define its **Joint Distribution Function**, denoted by $F_X$, as $F_X : \mathbb{R}^n \to \mathbb{R}$,

$$F_X(x_1, \ldots, x_n) = \mathbb{P}\left(\{X_1 \leq x_1\} \cap \{X_2 \leq x_2\} \cap \ldots \{X_n \leq x_n\}\right).$$

or $F_X(x_1, \ldots, x_n) = \mathbb{P}\left(\omega \in \Omega : \ X(\omega) \in ((-\infty, x_1) \times \ldots \times (-\infty, x_n))\right).$

- We can calculate $\mathbb{P}(X \in A)$ for any measurable subset $A \subset \mathbb{R}^n$.
- Properties
  1. $0 \leq F_X(x) \leq 1$ for any $x \in \mathbb{R}^n$.
  2. $\lim_{x_k \to -\infty} F_X(x) = 0$ for any $k = 1, \ldots, d$.
  3. $\lim_{x_1 \to +\infty, \ldots, x_n \to +\infty} F_X(x) = 1$.
  4. $F_X$ is left-continuous.

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# Continuous random vectors

## Definition

If the joint distribution function of a random vector $X$ is differentiable, we call it a **Continuous Random Vector** and we can define its **Joint Density Function**

$$f_X(x_1, \ldots, x_n) = \frac{\partial^n F}{\partial x_1 \partial x_2 \ldots \partial x_n}.$$

# Continuous random vectors

## Theorem

*We have*

- $f_X(x_1, \ldots, x_n) \geq 0$ *for any* $x = (x_1, \ldots, x_n) \in \mathbb{R}^n$.
- $\int_{R^n} f_X(x_1, \ldots, x_n) dx_1 dx_2 \ldots dx_n = 1.$
- *For any* $A \subset \mathbb{R}^n$,

$$\mathbb{P}(X \in A) = \int_A f_X(x_1, \ldots, x_n) dx_1 dx_2 \ldots dx_n.$$

- *For any index* $i$ *and* $k_i \in \mathbb{R}$,

$$\mathbb{P}(X_i \leq k_i) = \mathbb{P}(X \in (\mathbb{R} \times \ldots \times (-\infty, k_i) \times \ldots \times \mathbb{R})),$$

*for instance* $\mathbb{P}(X_1 \leq k_1) = \int_{-\infty}^{k_1} \int_{\mathbb{R}^{n-1}} f_X(x_1, \ldots, x_n) dx_1 dx_2 \ldots dx_n.$

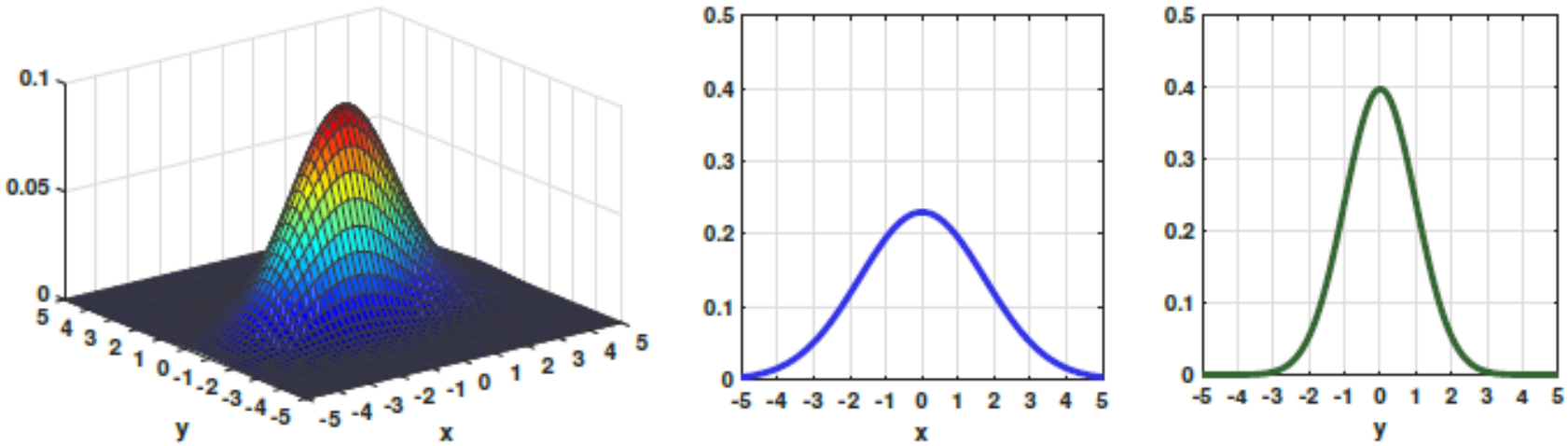



Figure: A two-dimensional Gaussian vector.

# Example

- Consider a joint density function

$$f(x,y) = \begin{cases} c & \text{if } (x,y) \in [0,3] \times [0,3] \\ 0 & \text{otherwise} \end{cases}$$

+ Find $c$.
+ Calculate the probability $\mathbb{P}(X + Y \leq 1)$.

$$1 = \int_{\mathbb{R}^2} f(x,y)dxdy = \int_0^3 \int_0^3 c dxdy = 9c \Rightarrow c = 1/9.$$

$$\mathbb{P}(X + Y \leq 1) = \int_{x+y\leq 1} f(x,y)dxdy = \int_{[0,3]^2 \cap \{x+y\leq 1\}} dxdy/9 = 1/18.$$

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- Consider a joint density function

$$f(x,y) = \begin{cases} ce^{-x}e^{-y} & \text{if } 0 \le y \le x < \infty \\ 0 & \text{otherwise} \end{cases}$$

Find $c$ and the marginal distribution of $X$ and $Y$.

- Hint: $1 = \int_{\mathbb{R}^2} f(x,y)dxdy$.

$$1 = \int_0^\infty dx \int_0^x ce^{-x}e^{-y}dy$$

$$= \int_0^\infty ce^{-x}(1-e^{-x})dx = c/2.$$

Hence $c = 2$.

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# Example

$$\begin{aligned}\mathbb{P}(X \leq t) &= \int_0^t dx \int_0^x 2e^{-x}e^{-y}dy \\ &= \int_0^t 2e^{-x}(1-e^{-x})dx = (1-2e^{-t}+e^{-2t}) \\ &=(1-e^{-t})^2.\end{aligned}$$

$$\begin{aligned}\mathbb{P}(Y \leq t) &= \int_t^\infty dx \int_0^t 2e^{-x}e^{-y}dy + \int_0^t dx \int_0^x 2e^{-x}e^{-y}dy \\ &=2(1-e^{-t})\int_t^\infty e^{-x}dx + \int_0^\infty 2e^{-x}(1-e^{-x})dx \\ &= = 2e^{-t}(1-e^{-t}) + (1-e^{-t})^2 = 1-e^{-2t}.\end{aligned}$$

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# Independent Random Variables

- Recall that: Two events $A$ and $B$ are independent if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B) \Leftrightarrow \mathbb{P}(A|B) = \mathbb{P}(A).$$

## Definition

Two random variables $X$ and $Y$ are **independent** if all possible couple of events deduce from them are independent, i.e. for any measurable subsets $A, b \subset \mathbb{R}$

$$\mathbb{P}(X \in A, Y \in B) = \mathbb{P}(X \in A)\mathbb{P}(Y \in B).$$

**Theorem**

*Two discrete random variables $X$ and $Y$ are independent if and only if*

$$\mathbb{P}(X = x_i, Y = y_j) = \mathbb{P}(X = x_i)\mathbb{P}(Y = y_j)\ \forall i, j.$$

**Theorem**

*For two continuous random variables $X$ and $Y$, the following three statements are equivalent*

- *They are independent.*
- $F_{X,Y}(x, y) = F_X(x)F_Y(y)$.
- $f_{X,Y}(x, y) = f_X(x)f_Y(y)$.

Check the variable-separability: $F_{X,Y}(x, y)$ (or $f_{X,Y}(x, y)$) $= h(x)g(y)$.

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- Check the independence of $X$ and $Y$.

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$$0 = \mathbb{P}(X = 2, Y = 2) \neq \mathbb{P}(X = 2)\mathbb{P}(Y = 2).$$

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# Example

- Consider a joint distribution function

$$F_{X,Y}(x,y) = (1-e^{-\lambda x})(1-e^{-\lambda y}), \quad x,y \geq 0.$$

- Find the joint density function
- Check the independence of $X$ and $Y$
- Find the distribution of $Z = X + Y$.

- Density function

$$f_{X,Y}(x,y) = \frac{\partial^2 F(x,y)}{\partial x \partial y} = \lambda^2 e^{-\lambda x} e^{-\lambda y}.$$

- Independence: YES!
- $\mathbb{P}(Z \leq z) = \int_0^z \int 0^{z-x} \lambda^2 e^{-\lambda x} e^{-\lambda y} dxdy.$

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# Expectation of Functions of Random Vectors

- Problem: Given a function $g : \mathbb{R}^2 \to \mathbb{R}$ and define a new random variable $Z = g(X, Y)$. How about the distribution and expectation of $Z$?

**Theorem**

- *For discrete random vector,*

$$\mathbb{E}[g(X,Y)] = \sum_{i,j} g(x_i, y_j)\mathbb{P}(X = x_i, Y = y_j) = \sum_{i,j} g(x_i, y_j)p_{i,j}.$$

- *For continuous random vector,*

$$\mathbb{E}[g(X,Y)] = \int_{\mathbb{R}^2} g(x,y)f_{X,Y}(x,y)dxdy.$$

-In general,

$$\mathbb{E}[g(X_1,\ldots,X_n)] = \int_{\mathbb{R}^n} g(x_1,\ldots,x_n)f(x_1,\ldots,x_n)dx_1\ldots dx_n.$$

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# Joint Expectation, Covariance

## Definition

- The expectation $\mathbb{E}[XY]$ is called the **joint expectation** of $X$ and $Y$.
- The **Covariance** of $X$ and $Y$ is defined as

$$\mathrm{Cov}(X,Y) = \mathbb{E}[(X - \mathbb{E}X)(Y - \mathbb{E}Y)] = \mathbb{E}[XY] - \mathbb{E}[X].\mathbb{E}[Y].$$

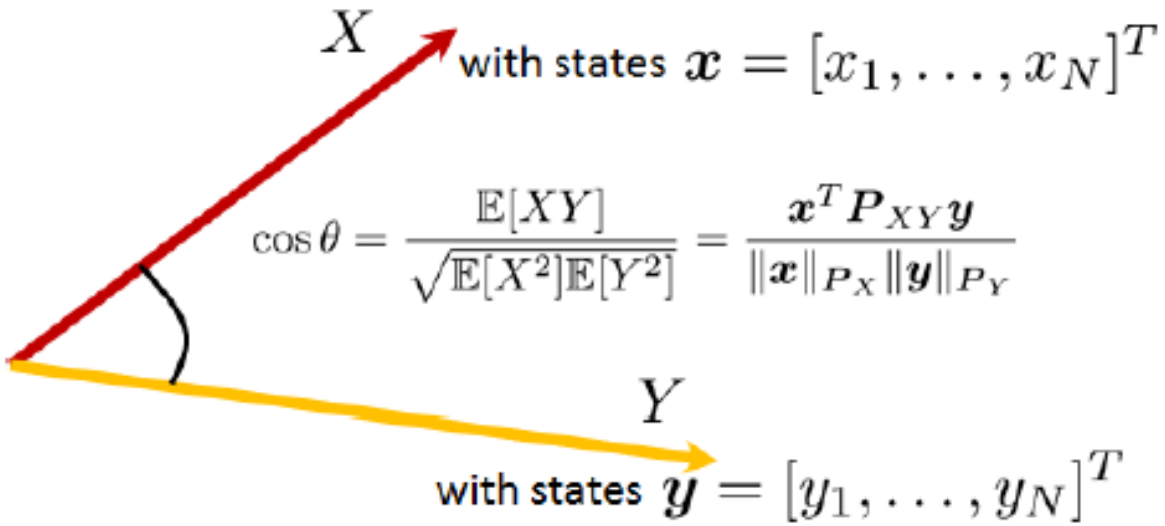
- The **Correlation coefficient** $\rho$ of $X$ and $Y$ is defined as

$$\rho(X,Y) = \frac{\mathrm{Cov}(X,Y)}{\sqrt{\mathrm{Var}(X)\mathrm{Var}(Y)}}.$$

If $\rho(X,Y) = 0$, we way that $X$ and $Y$ are **uncorrelated**.

- Remark: if $\mathbb{E}[X^2] < \infty$, we say that $X$ is square-integrable, denoted $X \in \mathcal{L}^2(\Omega)$.

# Geometric meaning of joint expectation




**Distributions**

$p_X(x)$ or $\boldsymbol{P}_X$

$p_{X,Y}(x, y)$ or $\boldsymbol{P}_{XY}$

$p_Y(y)$ or $\boldsymbol{P}_Y$

# Linearity of Expectation

## Theorem

*We have*

- $\mathbb{E}[X+Y] = E[X] + \mathbb{E}[Y]$.
- $Var(X+Y) = Var(X) + 2Cov(X,Y) + Var(Y)$.

$$-\mathbb{E}[X+Y] = \sum_{i,j}(x_i + y_j)\mathbb{P}(X = x_i, Y = y_j)$$

$$= \sum_i x_i \left( \sum_j \mathbb{P}(X = x_i, Y = y_j) \right) + \sum_j y_j \left( \sum_i \mathbb{P}(X = x_i, Y =$$

$$= \sum_i x_i \mathbb{P}(X = x_i) + \sum_j y_j \mathbb{P}(Y = y_j) = \mathbb{E}[X] + \mathbb{E}[Y].$$

$$-\mathrm{Var}(X+Y) = \mathbb{E}[((X+Y) - \mathbb{E}[X+Y])^2] = \mathbb{E}[((X - \mathbb{E}X) + (Y - \mathbb{E}Y))^2].$$

## Theorem

*We have*

$$(\mathbb{E}[XY])^2 \leq \mathbb{E}[X^2]\mathbb{E}[Y^2].$$

## Proof.

For any $t \in \mathbb{R}$,

$$\mathbb{E}[(tX + Y)^2] = t^2\mathbb{E}[X^2] + 2t\mathbb{E}[XY] + \mathbb{E}[Y^2] \geq 0.$$

Taking the delta of the quadratic function with respect to the variable $t$. □

## Theorem

$$|\rho(X,Y)| = \left| \frac{\mathbb{E}[(X-\mathbb{E}X)(Y-\mathbb{E}Y)]}{\sqrt{\mathbb{E}[(X-\mathbb{E}X)^2]\mathbb{E}[(Y-\mathbb{E}Y)^2]}} \right| \leq 1.$$

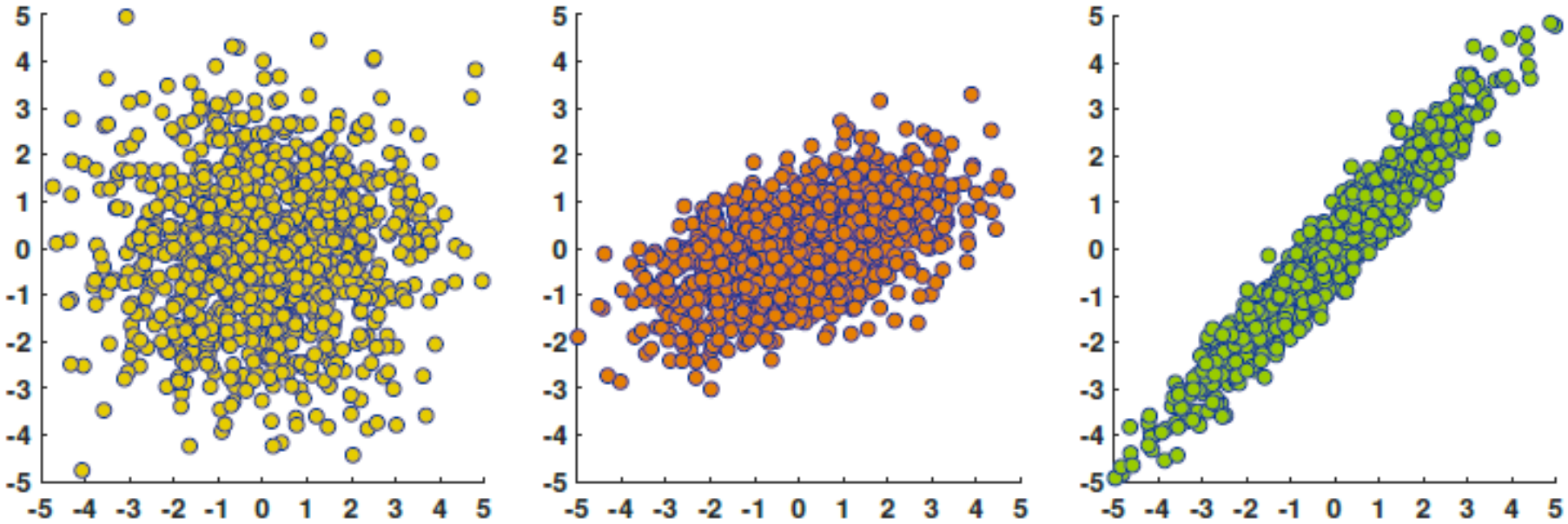



Figure: Visualization of correlated variables from data.

# Joint Expectation of Independent Random Variables

**Theorem**

*If $X$ and $Y$ are independent, then*

$$\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y].$$

$$\begin{aligned}
-\mathbb{E}[X+Y] &= \sum_{i,j} x_i y_j \mathbb{P}(X = x_i, Y = y_j) \\
&= \sum_i x_i \left( \sum_j x_i y_j \mathbb{P}(X = x_i)\mathbb{P}(Y = y_j) \right) \\
&= [\sum_i x_i \mathbb{P}(X = x_i)][\sum_j y_j \mathbb{P}(Y = y_j)] = \mathbb{E}[X]\mathbb{E}[Y].
\end{aligned}$$

Then **Independent** $\Rightarrow$ **Uncorrelated**, but the reverse is NOT TRUE!!!

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&= \sum_i x_i \left( \sum_j x_i y_j \mathbb{P}(X = x_i)\mathbb{P}(Y = y_j) \right) \\
&= [\sum_i x_i \mathbb{P}(X = x_i)][\sum_j y_j \mathbb{P}(Y = y_j)] = \mathbb{E}[X]\mathbb{E}[Y].
\end{aligned}$$

Then **Independent** $\Rightarrow$ **Uncorrelated**, but the reverse is NOT TRUE!!!

# Homework

+ Stanley Chan: Chapter5. 1-9, 11.
+ Ngo Hoang Long: Chapter2. 52, 53, 62.

Thank you!